

Clinical analysis report

Patient & case information

Doctor	yassin
Patient	koreaa
Patient code	1010
Age	25 years
Gender	Male
Case number	#179
Case date	19 Jun 2026
Created	19 Jun 2026 · 00:06

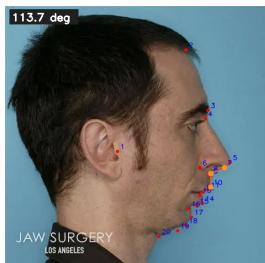
Medical History

Medical history has not been recorded.

Side view — measurement summary

Measurement	Value	Classification	Normal range	Clinical note
Nasiolabial angle	113.7°	Increased (>110°)	90° – 110°	Possible high columella or retruded maxilla. Class III cases may require surgical evaluation depending on full diagnosis.
Profile convexity angle	154.7°	Normal (151°–171°)	151° – 171°	Within normal range. No treatment implication from this angle alone.
Total facial convexity angle	124.1°	Decreased (<127°)	127° – 137°	Possible retruded chin or prominent nasal effect. May suggest Class II tendency; requires full orthodontic evaluation.
Mentolabial angle	91.1°	Decreased (<110°)	110° – 130°	Possible prominent chin, retroclined lower teeth, or mandibular retrusion. May require incisor correction, chin correction, or skeletal treatment depending on case.

Side view — measurement visualizations



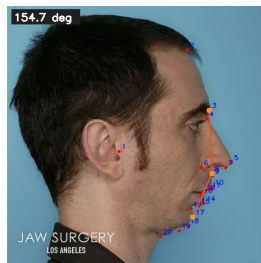
Nasiolabial angle **113.7°**

Normal: 90° – 110°

Increased (>110°)

Possible high columella or retruded maxilla.

Class III cases may require surgical evaluation depending on full diagnosis.



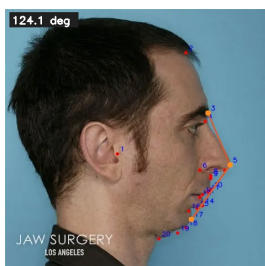
Profile convexity angle **154.7°**

Normal: 151° – 171°

Normal (151°–171°)

Within normal range.

No treatment implication from this angle alone.



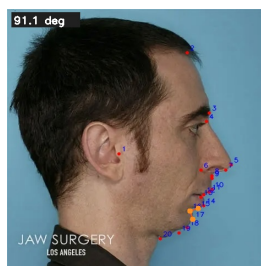
Total facial convexity angle **124.1°**

Normal: 127° – 137°

Decreased (<127°)

Possible retruded chin or prominent nasal effect.

May suggest Class II tendency; requires full orthodontic evaluation.



Mentolabial angle **91.1°**

Normal: 110° – 130°

Decreased (<110°)

Possible prominent chin, retroclined lower teeth, or mandibular retrusion.

May require incisor correction, chin correction, or skeletal treatment depending on case.

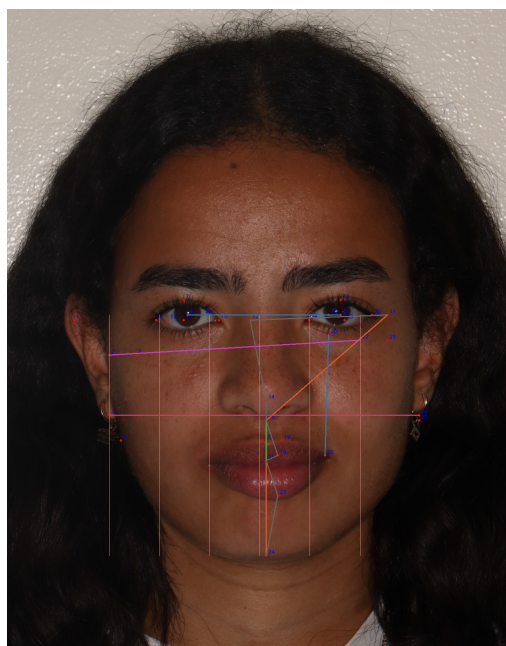
Frontal non-smile analysis

Frontal non-smile — measurement summary

Frontal non-smile — measurement summary (continued)

Measurement	Value	Interpretation
Middle vs lower facial third (proxy)	Middle 401.2 px, lower 337.0 px, ratio 1.191	Approximate vertical proportions; interpret with clinical context and growth pattern.
Upper / lower lip height ratio	0.347	Values far from ~1.0 may suggest disproportion; correlate with dental and soft-tissue exam.
Interpupillary vs commissure line	Interpupillary 0.0°, commissure 91.7°	Difference 91.7° exceeds typical cosmetic parallelism threshold (5.0°); clinical correlation suggested.
Facial midline (RMS deviation)	101.4 px	Midline deviation index is elevated; assess chin, nose, and dental midlines clinically.
Rule of fifths (approximate)	Ideal fifth width 125.6 px	Compare eye widths and intercanthal distance to one-fifth of bizygomatic width; rough aesthetic guide only.
Facial width (bizygomatic)	628.1 px	Useful for proportional checks; correlate with arch form and transverse skeletal pattern.
Mandibular width (bigonial)	774.0 px	Wide or narrow lower face relative to cheekbones informs transverse diagnosis.
Bizygomatic / bigonial ratio	0.811	Typical values often near ~1.2–1.4 in many samples; extremes suggest disproportion worth clinical review.
Facial index (height / width)	1.070	Longer narrower vs shorter wider facial form; integrate with vertical skeletal and growth assessment.
Interlabial gap (approx.)	91.7 px	Approximate soft-tissue opening at rest; verify with clinical lip posture exam.

Frontal visualization & interpretations



Annotated frontal overlay

Middle vs lower facial third (proxy)

Middle 401.2 px, lower 337.0 px, ratio 1.191

Approximate vertical proportions; interpret with clinical context and growth pattern.

Upper / lower lip height ratio

0.347

Values far from ~1.0 may suggest disproportion; correlate with dental and soft-tissue exam.

Interpupillary vs commissure line

Interpupillary 0.0°, commissure 91.7°

Difference 91.7° exceeds typical cosmetic parallelism threshold (5.0°); clinical correlation suggested.

Facial midline (RMS deviation)

101.4 px

Midline deviation index is elevated; assess chin, nose, and dental midlines clinically.

Rule of fifths (approximate)

Ideal fifth width 125.6 px

Compare eye widths and intercanthal distance to one-fifth of bizygomatic width; rough aesthetic guide only.

Facial width (bizygomatic)

628.1 px

Useful for proportional checks; correlate with arch form and transverse skeletal pattern.

Mandibular width (bigonial)

774.0 px

Wide or narrow lower face relative to cheekbones informs transverse diagnosis.

Bizygomatic / bigonial ratio

0.811

Typical values often near ~1.2–1.4 in many samples; extremes suggest disproportion worth clinical review.

Frontal interpretations (continued)

Facial index (height / width)

1.070

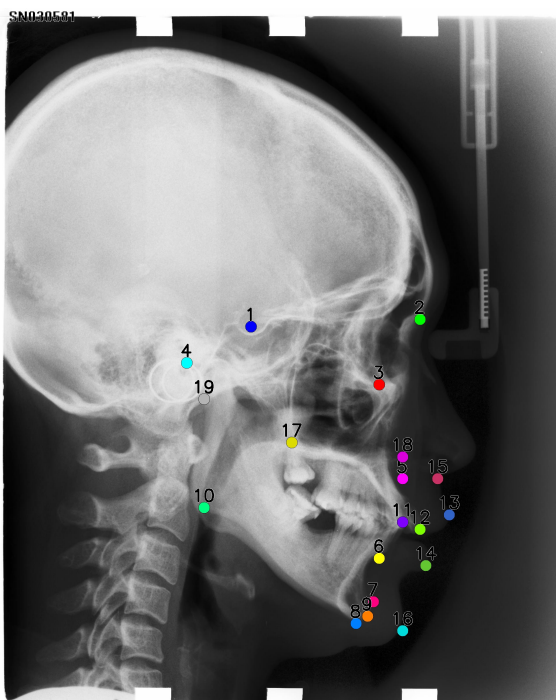
Longer narrower vs shorter wider facial form; integrate with vertical skeletal and growth assessment.

Interlabial gap (approx.)

91.7 px

Approximate soft-tissue opening at rest; verify with clinical lip posture exam.

X-Ray AI — Landmark Detection



19 Predicted Landmarks

Cephalometric landmarks predicted by HRNet AI from the X-ray image.

These landmarks are used to compute the profile angle measurements and the AI orthodontic diagnosis.

Annotated X-ray overlay — landmark positions are AI-predicted.

X-Ray AI — Primary Findings

Category	Finding	Conf %	Level	Status
Skeletal Class	Class II	94.6%	High	OK
Skeletal Mandible	Retrusion	95.3%	High	OK
Overjet	Large Overjet	98.7%	High	OK
Upper Lip Position	Normal	97.6%	High	OK

X-Ray AI — Supportive Findings

X-Ray AI — Supportive Findings (continued)

Category	Finding	Conf %	Level	Status
Dental Relationship	Class II	92.9%	High	■ Review
Overbite	Normal	94.3%	High	■ Review
Upper Incisor Inclination	Labioversion	81.3%	Medium	■ Review
Lower Incisor Inclination	Labioversion	98.2%	High	■ Review
Interincisal Angle	Small	96.5%	High	■ Review
Upper Incisal Display	Small	96.0%	High	■ Review
Profile Class	Plane	92.9%	High	■ Review

Pipeline: hrnet19_xgboost_11 | Landmarks: 19 | Features: 217 | AI decision-support only — must be reviewed by the orthodontist.

X-Ray AI Profile Measurements

Feature	Measurement	Value	Note
a_2_13_14	Nose–Upper Lip–Lower Lip Angle	146.7°	Relates the nasal landmark to upper and lower lip contour positions along the profile.
a_2_15_8	Nasal–Lip Root–Chin Angle	144.2°	Profile angle spanning from the nasal region through the upper lip root to the chin prominence.
a_7_15_8	Mid-Profile Support Angle	1.8°	Middle-face to chin profile angle; contributes to overall profile convexity assessment.
a_9_15_8	Chin Contour–Lip Root–Chin Angle	2.4°	Lower profile shape between two chin-area landmarks and the upper lip root.
a_2_13_8	Facial Profile Convexity Angle	130.8°	Full profile span from the nasal region through the upper lip to the chin point.
a_3_13_14	Subnasale–Upper Lip–Lower Lip Angle	126.9°	Sub-nasal point to lip relationship; may reflect lip prominence relative to nasal base.
a_13_14_8	Lip–Chin Angle	154.5°	Angle from the upper lip through the lower lip to the chin; relates to lip-chin balance.
a_15_8_9	Chin Prominence Angle	28.6°	Angle at the chin point between the upper lip root and the chin contour landmark.

These are AI landmark-derived profile features, not official cephalometric normative measurements.